Dynamic Modeling in Econometrics

Chapter 3

Foundational Knowledge for Time-Dependent Relationships

[Dynamic Modeling in Econometrics: Foundational Knowledge | Springer Nature Link]

Objectives

- Understand why dynamic models are essential in Econometrics
- Learn about Autoregressive (AR) models
- Master Distributed Lag (DL) and ARDL models
- Understand Moving Average (MA) and ARMA/ARIMA models
- Explore Vector Autoregression (VAR) models
- Learn about Vector Error Correction Models (VECM)

Importance of Dynamic Models

Definition: Models that capture time-dependent relationships between economic variables

Key Reasons:

- Temporal Dependencies: Past values influence current values
- Lagged Effects: Effects of changes take time to materialize
- Adjustment Dynamics: Variables adjust gradually to changes
- Policy Analysis: Understand delayed policy impacts
- Structural Breaks: Identify regime changes over time

Example: Interest rate changes affect GDP over several quarters, not immediately

Benefits of Dynamic Models

Benefit	**Explanation**
Temporal Dependencies	Captures how past values influence current values
Autoregressive Structure	Past dependent variable values predict current values
Adjustment Dynamics	Understand how variables adjust to changes
Speed of Adjustment	How quickly variables return to equilibrium
Cointegration	Model both short and long-term relationships
Policy Impact Assessment	Delayed effects of policy interventions
Endogeneity Control	Lagged dependent variables help control endogeneity
Structural Break Detection	Identify regime changes in data

Types of Dynamic Models

Model	Description
AR	Uses past values of dependent variable
DL	Uses lagged values of independent variables
ARDL	Combines AR and DL
MA	Current value depends on past errors
ARMA	Combines AR and MA for stationary series
ARIMA	ARMA with differencing for non-stationary series
VAR	Multiple time series relationships
VECM	VAR with cointegration for long-term equilibrium

Autoregressive (AR) Models

Definition: Regress a variable on one or more of its lagged values

AR(k) Model:

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \ldots + \beta_k Y_{t-k} + \epsilon_t$$

AR(1) Model:

$$Y_t = \alpha + \beta_1 Y_{t-1} + \epsilon_t$$

Key Requirements:

- Stationary Data: Mean, variance do not change over time
- Persistence: Variables influenced by own past values

Use Cases:

- Economic forecasting
- Signal processing
- Building block for ARIMA, VAR models

Distributed Lag (DL) Models

Definition: Effect of regressor X on Y occurs over time, not all at once

Finite DL (FDL) Model:

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \ldots + \beta_k X_{t-k} + \epsilon_t$$

Infinite DL (IDL) Model:

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \ldots + \infty + \epsilon_t$$

Key Terms:

- Short-run impact: β_0
- Lag multipliers: $\beta_1, \beta_2, \ldots, \beta_k$
- Long-run impact: $\sum_{i=0}^{k} \beta_i$

Koyck Transformation

The Problem:

- Infinite number of parameters to estimate
- Multicollinearity between lagged values

Koyck's Solution: Assume coefficients decline geometrically:

$$\beta_k = \beta_0 \lambda^k, \qquad 0 < \lambda < 1$$

Results:

- λ near 1: Slow decline, distant past matters
- λ near 0: Fast decline, only recent past matters

Transformed Model:

$$Y_t = \alpha(1-\lambda) + \beta_0 X_t + \lambda Y_{t-1} + (\epsilon_t - \lambda\epsilon_{t-1})$$

Only 3 parameters to estimate!

Koyck Model - Interpretation

Original IDL:

$$Y_t = \alpha + \beta_0 X_t + \beta_0 \lambda X_{t-1} + \beta_0 \lambda^2 X_{t-2} + \ldots + \infty + \epsilon_t$$

After Koyck Transformation:

$$Y_t = \alpha(1-\lambda) + \beta_0 X_t + \lambda Y_{t-1} + (\epsilon_t - \lambda \epsilon_{t-1})$$

Impact Multipliers:

- Short-run impact: β_0
- Long-run impact: $\frac{\beta_0}{1-\lambda}$
- Long-run > Short-run (since $0 < \lambda < 1$)

Caution:

- Error term is autocorrelated
- May need HAC standard errors

Limitations of DL Models

Too Few Lags:

- Miss long-term dependencies
- Poor model fit
- Residual autocorrelation

Too Many Lags:

- Overfitting: Captures noise, not patterns
- Multicollinearity: Correlated lagged variables
- Reduced Observations: Lose data points
- Unstable Estimates: Coefficients become unreliable

Solution: Use information criteria (AIC, BIC) to select optimal lag length

Autoregressive Distributed Lag (ARDL)

Definition: Combines AR and DL models

General ARDL(p,q) Model:

$$Y_t = \alpha + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=0}^{q} \beta_j X_{t-j} + \varepsilon_t$$

- p: Order of autoregressive part
- q: Order of distributed lag part

With Trend and Seasonality:

$$Y_t = \alpha + \gamma t + \sum_{s=1}^{S} \delta_s D_s + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=0}^{q} \beta_j X_{t-j} + \varepsilon_t$$

AR, DL, ARDL Comparison

Feature	AR Model	DL Model	ARDL Model
Focus	Lagged values of Y_t	Lagged values of X_t	Combines both
Stationarity	Requires stationary Y_t	Requires stationary X_t	Allows mix of I(0) and I(1)
Flexibility	Captures autocorrelation	Captures lagged effects	Captures both
Complexity	Simpler	Simpler	Comprehensive

Koyck vs ARDL

Aspect	Koyck Transformation	ARDL
Model Structure	Geometric decay of past effects	Flexible lag structure
Formula	$Y_t = \alpha(1-\lambda) + \beta_0 X_t + \lambda Y_{t-1} + \nu_t$	$Y_t = \alpha + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=0}^{q} \beta_j X_{t-j} + \varepsilon_t$
Error Terms	Includes lagged error (autocorrelation)	Flexible error structure
Dynamics	Focuses on decay effect	Models both short and long-run
Use Case	Simpler, imposes structure	More robust, flexible

Moving Average (MA) Models

Definition: Combines AR and MA processes

ARMA(p,q) Model:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t$$

Components:

- AR part: Own past values ($Y_{t-1}, Y_{t-2}, \ldots$)
- MA part: Past error terms ($\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$)

Requirements:

- Stationary time series
- Mean zero (or constant mean)

Use Case: Forecasting stationary series

ARIMA Models

Definition: ARMA with differencing for non-stationary series

Model Equation:

$$\Delta^d Y_t = \phi_1 Y_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

Components:

- AR(p): Autoregressive part
- I(d): Integrated (differencing) part
- MA(q): Moving Average part

Notations:

- ARIMA(1,0,0) = AR(1)
- ARIMA(0,0,1) = MA(1)
- ARIMA(1,1,1) = ARMA(1,1) with one differencing

Purpose:

- Make non-stationary data stationary
- Remove trends and seasonality

Vector Autoregression (VAR)

Motivation: Most economic variables are interdependent

Limitations of Univariate Models:

- Predict single variable only
- Ignore cross-variable relationships
- Cannot capture feedback mechanisms

VAR Advantages:

- Models multiple variables simultaneously
- Captures dynamic interactions
- No strong theoretical restrictions

Primary Objectives:

- Characterize macroeconomic data
- Forecast economic conditions
- Understand underlying structure
- Advise policymakers

VAR Model Structure

Vector Form:

$$Y_t = \begin{bmatrix} Y_{1t} \\ Y_{2t} \\ \vdots \\ Y_{nt} \end{bmatrix}, \epsilon_t = \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \vdots \\ \epsilon_{nt} \end{bmatrix}$$

VAR(p) Model:

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \ldots + \beta_n Y_{t-n} + \epsilon_t$$

Components:

- α: Vector of constants
- β_i: Matrix of coefficients for i-th lag
- ϵ_t: Vector of error terms

VAR(2)

VAR(2) with Two Variables

Matrix Form:

$$\begin{bmatrix} Y_{1t} \\ Y_{2t} \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \beta_{11,1} & \beta_{12,1} \\ \beta_{21,1} & \beta_{22,1} \end{bmatrix} \begin{bmatrix} Y_{1,t-1} \\ Y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \beta_{11,2} & \beta_{12,2} \\ \beta_{21,2} & \beta_{22,2} \end{bmatrix} \begin{bmatrix} Y_{1,t-2} \\ Y_{2,t-2} \end{bmatrix} + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}$$

Interpretation:

- Each variable depends on its own past values AND past values of other variables
- Bidirectional relationships captured
- Simultaneous modeling of multiple variables

Reduced Form VAR

Three Variables:

- Unemployment Rate
- Inflation
- Interest Rate

Equation for Unemployment:

$$Unemp_t = \beta_{10} + \beta_{11}Unemp_{t-1} + \beta_{12}Unemp_{t-2} + \gamma_{11}Inf_{t-1} + \gamma_{12}Inf_{t-2} + \phi_{11}Rate_{t-1} + \phi_{12}Rate_{t-2} + \epsilon_{1t}$$

Key Features:

- Only lagged values of other variables
- No contemporaneous relationships
- Correlated errors across equations
- Simpler to estimate (OLS)

Structural VAR (SVAR)

Key Difference: Explicitly models contemporaneous relationships

Structural Form:

$$B\begin{bmatrix}Y_{1t}\\Y_{2t}\end{bmatrix}=\begin{bmatrix}\alpha_1\\\alpha_2\end{bmatrix}+\begin{bmatrix}\beta_{11,1} & \beta_{12,1}\\\beta_{21,1} & \beta_{22,1}\end{bmatrix}\begin{bmatrix}Y_{1,t-1}\\Y_{2,t-1}\end{bmatrix}+\begin{bmatrix}\beta_{11,2} & \beta_{12,2}\\\beta_{21,2} & \beta_{22,2}\end{bmatrix}\begin{bmatrix}Y_{1,t-2}\\Y_{2,t-2}\end{bmatrix}+\begin{bmatrix}\epsilon_{1t}\\\epsilon_{2t}\end{bmatrix}$$

Relationship:

$$B=\begin{pmatrix}1 & -b_{12}\\-b_{21} & 1\end{pmatrix}$$

Link to Reduced Form:

$$u_t=B^{-1}\epsilon_t$$

VAR vs SVAR Comparison

Aspect	**Reduced Form VAR**	**Structural VAR (SVAR)**
Error Terms	Correlated (u_t)	Uncorrelated (ϵ_t)
Contemporaneous Relationships	Not modeled	Explicitly modeled
Causal Interpretation	Lacks causation	Provides causal relationships
Identification	No restrictions needed	Requires theoretical restrictions
Estimation	Simpler (OLS)	More complex
Use Case	Forecasting, describing dynamics	Policy analysis, structural inference

VAR Applications

Impulse Response Functions (IRF):

- Show how shocks to one variable affect others over time
- Understand dynamic relationships and feedback mechanisms

Forecast Error Variance Decomposition (FEVD):

- Proportion of variance explained by each variable
- Understand relative importance of different variables

Policy Analysis:

- Analyze monetary policy shocks
- Study fiscal policy impacts
- Understand international spillovers

Vector Error Correction Model (VECM)

Problem: VAR assumes all variables are stationary

Issue with Non-Stationary Variables:

- Spurious regression
- High R^2 with meaningless relationships
- Invalid inference

Cointegration Solution:

- Even if individual series are non-stationary
- They may move together in the long run
- Linear combination is stationary

VECM Purpose:

- Handle non-stationary cointegrated variables
- Separate long-term equilibrium from short-term dynamics
- Include error correction mechanism

Cointegration

Simple Regression: $Y_t = \beta_0 + \beta_1 X_t + \epsilon_t$
Error Term: $\epsilon_t = Y_t - \beta_0 - \beta_1 X_t$

If both X_t and Y_t are non-stationary:

- ϵ_t is also non-stationary
- Violates assumption: $\epsilon_t \sim N(0, 1)$

If ϵ_t is stationary:

- Cointegration exists!
- Variables have a long-run relationship
- Regression is NOT spurious

Error Correction Model (ECM)

Start with VAR(1):

$$Y_t = \beta_0 + \beta_1 X_t + \alpha_2 Y_{t-1} + \beta_2 X_{t-1} + \epsilon_t$$

Rearrange:

$$\Delta Y_t = \beta_1 \Delta X_t + \varphi(Y_{t-1} - \theta_1 - \theta_2 X_{t-1}) + \epsilon_t$$

Where:

- $\varphi = (\alpha_2 - 1)$ (error correction parameter)
- $\theta_1 = \frac{\beta_0}{1-\alpha_2}$
- $\theta_2 = \frac{\beta_1+\beta_2}{1-\alpha_2}$

Interpretation:

- ΔX_t: Short-run dynamics
- $\varphi(\ldots)$: Error correction mechanism

ECM - Two Equations

For Y and X:

- $\Delta Y_t = \alpha_1 + \beta_{11}\Delta Y_{t-1} + \beta_{12}\Delta X_{t-1} + \varphi_Y(Y_{t-1} - \theta_1 - \theta_2 X_{t-1}) + \epsilon_{Y,t}$
- $\Delta X_t = \alpha_2 + \beta_{21}\Delta Y_{t-1} + \beta_{22}\Delta X_{t-1} + \varphi_X(Y_{t-1} - \theta_1 - \theta_2 X_{t-1}) + \epsilon_{X,t}$

Components:

- φ_Y, φ_X: Error correction coefficients
- Measure speed of adjustment to equilibrium
- Should be negative for stability

VAR vs VECM vs ARDL

Aspect	**VAR**	**VECM**	**ARDL**
Purpose	Model dynamic relationships	Long-term equilibrium + short-term dynamics	Single equation with both dynamics
Stationarity	All variables stationary	Variables cointegrated	Mix of I(0) and I(1)
Long-term Equilibrium	Not modeled	Explicitly modeled	Can be derived
Equation System	Multiple equations	Multiple equations	Single equation
Use When	Short-term focus, stationary data	Cointegrated variables	Mixed integration orders

VAR Model Specification

Information Criteria:

- AIC (Akaike Information Criterion)
- BIC (Bayesian Information Criterion)
- HQ (Hannan-Quinn Criterion)
- FPE (Final Prediction Error)

General Approach:

- Estimate VAR for different lag lengths ($p = 1, 2, 3, \ldots$)
- Choose p that minimizes information criteria
- Typically symmetric lags (same for all variables)

Trade-off:

- Too few lags: Miss dynamics
- Too many lags: Overfitting, multicollinearity

Questions & Discussion

Review Questions:

- What is the difference between AR and DL models?
- When would you use a VAR model vs VECM?
- What does the Koyck transformation do?
- How do you select the lag length for a VAR?
- What is cointegration and why is it important?

Discussion Points:

- Advantages and limitations of dynamic models
- When to use structural vs reduced form models
- Practical applications in policy analysis

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